

Summative Assignment

Module code and title	COMP3607 Recommender Systems
Academic year	2025-26
Coursework title	RecSys Assignment
Coursework credits	10 credits
% of module's final mark	100%
Lecturer	Suncica Hadzidedic
Submission date*	Thursday, 18 December, 2025 14:00
Estimated hours of work	20
Submission method	Gradescope

Additional coursework files	N/A
Required submission items and formats	Submit via Gradescope : • Video - presentation and demonstration (any format supported) • Python code (.zip file)

* This is the deadline for all submissions except where an approved extension is in place. For assessments carried out in weekly practicals or other scheduled sessions, the date shown will be the Monday of the week in which the assessments take place.

Late submissions received within 5 working days of the deadline will be capped at 40%.

Late submissions received later than 5 days after the deadline will receive a mark of 0.

It is your responsibility to check that your submission has uploaded successfully and obtain a submission receipt.

Your work must be done by yourself (or your group, if there is an assigned groupwork component) and comply with the university rules about plagiarism and collusion. Students suspected of plagiarism, either of published or unpublished sources, including the work of other students, or of collusion will be dealt with according to University guidelines (<https://www.dur.ac.uk/learningandteaching.handbook/6/2/4/>).

COMP3607 Coursework:

Designing and Developing Personalised Recommender Systems

I. Requirements

For this coursework, you are required to design, develop and evaluate two recommender systems (RSs):

1. **RS1:** A **personalised, conventional RS** (e.g., content-based filtering with TFIDF).
2. **RS2:** An **advanced RS**, that extends the conventional RS1, by using **more advanced or state-of-the-art methods** (e.g., deep learning, graph neural networks, LLMs).

Having implemented and evaluated your two systems, you will submit:

1. A **10-minute video** (with audio) comprising a recorded presentation (e.g., ppt) explaining your systems' methods, implementation (code), evaluation results, and a demonstration of user interaction with your RSs.
2. A .zip file containing all the **Python code** used to implement and evaluate your systems (use Python 3.x).

N.B. Marks are awarded based on your **video submission**. Your code submission will be used **only** as evidence of the implementation and evaluation of the two RSs. Therefore, you must ensure that your video submission covers **all the points** listed in Section 1.5 *Video submission*, including a demonstration of the code within the video. To reiterate, **marks will NOT be awarded for work presented in the code submission**. However, videos received without an accompanying code submission will not be marked.

1.1 Dataset

- You can choose any **domain** and **dataset** for your RS.
- Randomly sample 100,000 (100K) cases from the original dataset. (*Note: You can use larger samples, however, additional marks will not be given for this.*)
- You must use the same 100K dataset for both RS1 and RS2.
- Describe the dataset and explain the data preparation methods in the video.
- A list of some publicly available datasets is provided in Section 3. You are also free to use other datasets or synthetically generated data.

1.2 RS design and development

- You should research and select the most **appropriate/suitable methods** for your RSs, including those for: user profile learning, rating (usage) prediction, evaluation metrics, etc.
 - a. RS1: You can choose any conventional RS method (e.g., CF with matrix factorization, hybrid of two conventional techniques, etc.).

- b. RS2: This RS will build upon RS1, by replacing the traditional method with more advanced or state-of-the-art methods. These methods can be applied to, e.g.: user preference/rating prediction; processing multimodal data; data augmentation; explanations of recommendations; etc.
- In the video presentation, you must **describe** and **justify** the methods you used. Justifications should address the selected methods' relevance to and alignment with the purpose, application domain and data used for the implemented recommender systems. Use relevant literature to support your justifications.
- Responsible AI considerations: Outline and discuss any ethical concerns stemming from the methods you chose (e.g., data, models, output) for both RS1 and RS2.

1.3 Evaluation

Evaluate the performance of your RSs by carrying out an **offline experiment**, as follows:

- Metric 1: Choose one **metric** from the following: accuracy of **rating** predictions, accuracy of **ranking**.
- Metric 2: And choose one more **metric** from the following: serendipity, explainability, fairness.
- Use the two metrics you have chosen to evaluate the performance of your two RSs.
- **Justify** in the video presentation why the selected metrics are appropriate for the purpose of the RS and domain of application, and include supporting references.
- **Compare** the performance of RS1 against RS2 on **both** evaluation metrics you have chosen. You must present the comparison in the video submission, and make sure to:
 - Clearly present (e.g., in tables or graphs) and interpret the results for metric 1 and metric 2, for both RS1 and RS2.
 - Demonstrate the code that shows you have implemented and applied the evaluation metrics.
 - Compare the two RSs and draw conclusions about RS2 performance.

1.4 User interface

The UI should be a **command-line** interface for both RS versions. You must implement a command-line interface for both RS1 and RS2.

Consider UI design principles, including the following:

- *Input UI*: How does the system recognise an active user?
- *Output UI*: Number of recommendations; style of output presentation.

N.B. Do **not** develop graphical, web-based interfaces. These will not be marked.

1.5 Video submission

The video should showcase your overall work. It must be up to 10 minutes (and not longer) and include both video and audio recording. (N.B. *Videos without an audio recording will have a penalty.*)

The video will consist of two main parts - presentation and RS demo. You are free to distribute the 10 minutes between the two parts as you see fit.

i. **Presentation slides** (e.g., Power Point) must cover the following content:

- Introduction
 - Domain of application
 - Purpose/Aim
- Methods and their justification
 - Dataset description
 - Data preparation – separately for RS1 and RS2
 - Recommendation techniques/algorithms – separately for RS1 and RS2
- Evaluation – separately for RS1 and RS2
 - Evaluation metrics (in equations) and justification
 - Evaluation results (in tables or graphs)
- Discussion on ethical concerns – separately for RS1 and RS2
- RS1 vs RS2 comparisons and conclusions (performance metrics and ethics)
- References in IEEE style

ii. **RS demo**

- Demonstrate RS1 and RS2 separately, on each of the following points:
 - Choose **two** users from your dataset. For those two users run a live demonstration of each RS. Show the UI and personalised recommendations that RS1 generates for each user, and do the same for RS2.
 - Go through the relevant parts of your **code submission** and explain the main:
 - techniques/algorithms used
 - evaluation metrics and performance results.

1.6 Code submission

You must implement your RSs in Python 3.x. Your code submission must include:

- All the code necessary to produce the implementation and evaluation of the two RSs which you described in the video.
- A **requirements.txt** file explicitly recording the versions of all modules needed to run all your code and the version of Python (3.x) you have used.
- A **readme.txt** file containing instructions how to run your code, and noting any required dependencies other than those listed in requirements.txt.
- A Python 3.x program **run_eval.py** that when run using “python run_eval.py” evaluates both RS1 and RS2, and prints a summary of the evaluation results.

II. Marking Scheme

The distribution of coursework marks is presented in the table below.

RS1. Conventional RS methods: dataset, design and development, UI demo	35
RS1. Evaluation: evaluation metrics, results, ethics	15
RS2. Advanced RS methods: design and development, UI demo	20
RS2. Evaluation: evaluation results, ethics, comparison	15
Presentation: format, structure, clarity, content	10
Participation: formative assignment, peer-review, weekly tasks	5
TOTAL /100	

III. Public datasets

These are some of the available datasets, others are listed on our Ultra page: [Teaching Material->Datasets](#). You can choose any appropriate dataset, not just those listed below.

Dataset	Link	Description
Yelp	https://www.yelp.com/dataset	user reviews of different businesses and services in a specific location
ReDial	https://redialdata.github.io/website/	Dialogues of users recommending movies to each other
LDOS-CoMoDa	https://www.lucami.org/en/research/ldos-comoda-dataset/	context-aware movie data
Million Song	http://millionsongdataset.com/	music and context data
Last.fm	http://millionsongdataset.com/lastfm/	song tag and song similarity
GroupLens	https://grouplens.org/datasets/	movies, books, personality-aware data
Inspired	https://github.com/sweetpeach/Inspired	1,001 human-human dialogs for movie recommendation
UC San Diego	https://cseweb.ucsd.edu/~jmcauley/datasets.html	RS datasets for Amazon reviews, Goodreads, clothing data, etc.
KuaiRand	https://kuairand.com/	Videos - unbiased sequential recommendation
KuaiRec	https://kuaiREC.com/	Kuaishou videos - fully observed user-item interaction matrix
Criteo Click Logs	https://ailab.criteo.com/download-criteo-1tb-click-logs-dataset/	Ads - feature values and click feedback
Epinions	https://www.cse.msu.edu/~tangjili/datasetcode/truststudy.htm	User profile, ratings and trust relations
Ciao	https://paperswithcode.com/dataset/ciao	Trust in RS: rating information of users given to items, and also contain item category information
Douban	https://paperswithcode.com/dataset/douban	social network: user review and recommendation services for movies, books, and music
Taobao	https://www.comp.hkbu.edu.hk/~lichen/download/TaoBao_Serendipity_Dataset.html	e-commerce, user perception of recommendation, curiosity and personality

IV. FAQs

FAQs from prior cohorts and your Q&A are available on the [discussion board](#) on the RS Ultra page.